

INVESTIGATION REPORT

Issue 993: Does CV32E40X fetch its first instruction from boot+4?

openhwgroup/cv32e40x

Document	RTL-TRIAGE-CV32E40X-993
Disposition	Not reproduced
Record date	2026-09-10
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40x#993
Issue filed	2026-05-03
Issue type	Bug report from an integrator or user
Affects	cv32e40x
Configuration	X_EXT=0 (default), RTL master
Disposition	Not reproduced
Severity	Not assessed
Agreement	No independent human record was available for comparison.
Evidence	5 cited, 3 present, 2 not retained

Disposition meaning. The reported behaviour did not occur under the conditions investigated. This is not a statement that no defect exists.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The reported behaviour (first accepted instruction fetch after reset at boot_addr_i + 4) does not occur at the cv32e40x_core boundary under a quiet environment. With OBI handshake legality, reset released once, boot_addr_i stable, and no debug request / interrupt / bus error until the first fetch is accepted, an unbounded proof (abc pdr) plus a bounded check over 30 cycles (the first fetch is accepted in cycle 3) show the first accepted instr_addr_o equals {boot_addr_i[31:2],2'b00}, and a cover for 'first accepted fetch == boot+4' is unreachable in that window. Without the quiet-environment assumption the solver does produce a counterexample, but it is an environment artifact: it asserts debug_req_i while the core is in RESET, so the core enters DEBUG_TAKEN and its first fetch goes to dm_halt_addr_i (a free input the solver set to boot+4). The most likely explanation of the reporter's observation is therefore something in their integration (a debug halt or interrupt at boot, or a monitor sampling instr_addr_o one cycle after gnt), not the core. Recommended: ask the reporter for debug_req_i/irq_i/dm_halt_addr_i values at reset and the exact cycle of gnt relative to instr_addr_o.

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Checker used

checker_993_quiet.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/prove_quiet/sby.log	present	abc pdr unbounded PASS (quiet environment): first accepted fetch after reset == boot address, for every input sequence
E2	proof	formal/bmc_quiet/sby.log	present	BMC depth 30, quiet environment: assertion 'first accepted fetch == boot address' PASS
E3	proof	formal/cover_quiet/sby.log	present	cover 'first accepted fetch' REACHED at step 3; cover 'first accepted fetch == boot+4' UNREACHED to depth 30
E4	counterexample	formal/bmc_fixedboot/job/engine_0/trace.vcd	not kept	Unconstrained-debug counterexample: debug_req_i during RESET -> DEBUG_TAKEN -> first fetch to dm_halt_addr_i. Explains how a 'wrong first fetch' can be observed; rendered in cex_env_artifact.png
E5	counterexample	formal/cover_quiet/job/engine_0/trace0.vcd	not kept	witness of the normal boot: fetch_branch to boot at cycle 2, accepted request at boot address cycle 3; cover_quiet_first_fetch.png

Paths are relative to runs/e40x-993. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.

Artefact(s) E4, E5 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

7 Limitations and scope

- Absence of reproduction is not proof of absence of a defect. A different configuration, stimulus or tool version may still expose the reported behaviour.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory	runs/e40x-993
Title source	Title and configuration read back from reports/issues/cv32e40x_issue_993.pdf, the previously published report for this issue.
Inputs shown	title, body
Human record	open issue, no maintainer response yet (as of 2026-09-10)
Assessed confidence	0.9. This is the investigator's own estimate. It is not a measurement and not a reliability figure.
Iterations	5

Generated by reporting/scripts/issue_document.py from runs/e40x-993/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.